

From Embeddings to Clusters: A Bilingual NLP Approach to Disaster Event Detection

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Abstract—Natural hazards are increasingly frequent and severe, affecting millions globally and placing immense pressure on crisis response systems. For countries where resources are limited and language barriers exist, the need for effective, real-time disaster detection from local data sources is crucial for minimizing harm and maximizing response efficiency. In Bengali-speaking regions, one of the world’s most vulnerable areas to climate-induced disasters, a lack of automated tools for disaster event detection presents significant challenges. This research focuses on building a robust framework for disaster event detection by leveraging advanced Natural Language Processing (NLP) techniques. Our approach involves creating a labeled dataset, generating contextual embeddings, and utilizing clustering mechanisms for effective similarity matching and classification. This approach shows promising results where the highest achieved F1-score is @@@@. Through this research, we aim to contribute a reliable, scalable solution for natural disaster detection that could aid governmental and non-governmental organizations in crisis management, ultimately fostering resilience and safety in affected communities.

Index Terms—Contextual embeddings, Pre-trained models, Clustering mechanisms, Natural text analysis, Natural Disaster Detection

I. INTRODUCTION

The increasing frequency and intensity of natural disasters have emerged as one of the most pressing global challenges of the 21st century. Bangladesh, one of the most disaster-prone countries in the world, faces frequent natural calamities such as riverine floods, earthquakes, hurricanes, and droughts that cause large-scale human and economic losses each year. In 2020, Cyclone Amphan swept across the southern coast, destroying thousands of homes and displacing millions. Two years later, the 2022 Sylhet floods submerged vast areas of northeastern Bangladesh, leaving communities stranded for days without adequate relief. What made both disasters more tragic was not the lack of information — hundreds of news articles, social media posts, and eyewitness reports were available online in real time — but rather the absence of an automated system capable of identifying and prioritizing these critical pieces of information. Despite the abundance of textual data on digital platforms (Imran et al., 2020; Alam et al., 2021) [12], the national response systems struggled to detect the severity and scope of these events quickly enough, resulting in delayed humanitarian actions and preventable losses.

This abundance of unstructured textual information presents a unique opportunity: when analyzed effectively, it can provide early indicators of emerging disasters, complementing traditional monitoring systems that rely on manual information extraction or English-language sources.

Unlike numerical sensor data, text often contains rich contextual cues—locations, damage descriptions, and human sentiments—that reveal on-the-ground realities more comprehensively. While event detection has been widely explored in high-resource languages such as English (Chen et al., 2017; Li et al., 2020) [13], Bengali remains underrepresented, with limited annotated data and computational resources (Islam et al., 2022) [14].

To bridge this gap, the present study introduces a novel framework for disaster event detection in Bengali and English text using semantic embedding and clustering-based analysis. The approach employs multilingual Bidirectional Encoder Representations from Transformers (BERT) for contextual embedding and Facebook AI Similarity Search (FAISS) for clustering — a combination not previously applied to Bengali text. The system transforms unstructured news data into semantically meaningful vector representations and clusters them according to disaster-related contexts, enabling automatic identification of emerging events with high precision, even without explicit keywords. By demonstrating the potential of deep semantic models in a low-resource setting, this work lays the foundation for real-time, text-based disaster monitoring and contributes to the advancement of data-driven humanitarian response systems.

II. RELATED WORKS

The detection of natural disaster events using textual data has been a subject of significant interest in recent years, particularly with the increasing availability of data from

social media and online news platforms. Prior research has explored several key areas leveraging advancements in machine learning, natural language processing (NLP), and data clustering techniques.

Social media platforms like Twitter and Facebook are widely recognized as valuable sources for real-time disaster event detection. For instance, Sakaki et al. (2010) [6] demonstrated an earthquake detection system using Twitter data, employing keyword-based filtering and semantic analysis. Similarly, CrisisNLP by Imran et al. [7] highlighted the potential of multilingual datasets for addressing diverse populations. The study by Sánchez et al. (2023) [8] proposed a cross-lingual and cross-domain classification framework that leverages labeled data from high-resource languages to classify crisis events in low-resource languages. Their approach achieved promising results, emphasizing the potential of multilingual and domain-transfer methods in NLP.

Empirical studies have proved that vectorization of texts or word embeddings like Word2Vec of Mikolov et al. (2013) [9] and contextual embeddings such as BERT (Devlin et al., 2019) [10] have significantly enhanced text analysis, capturing semantic nuances in datasets. These approaches form the foundation for effective classification and clustering tasks from natural texts. Facebook AI Similarity Search (FAISS) has been used for high-dimensional data clustering and retrieval, enabling efficient handling of large datasets, as shown by Johnson et al. (2017) [11]. This tool is particularly relevant for processing large-scale multilingual datasets, as demonstrated in this study.

While these studies have made significant strides in disaster detection using textual data, limitations remain, particularly in addressing underrepresented languages like Bengali and in leveraging clustering for manually annotated datasets. Building on these foundations, this study integrates clustering, embeddings, and multilingual analysis to advance disaster detection. The details of this methodology will be discussed in the following sections.

III. METHODS AND PROCEDURES

A. Dataset Structure

Dataset is crucial for AI models as dataset structure plays a prominent role in training a machine learning system and can significantly affect its outcome [2]. The proposed dataset is designed to facilitate efficient and accurate detection of natural disaster-related content in multilingual text data. It is organized into two primary columns:

1. **Sample:** This column contains the text data, consisting of sentences extracted from various sources. The sentences are in English and Bengali, representing a diverse linguistic dataset.

2. **Tag:** This column labels each sentence with one of the 18 predefined [1] natural disaster categories. Also there are additionally two categories: Disaster and Non-Disaster to generalize ambiguous data.

Sample	Tag
Images filmed from a refuge close to the incident show snow and rock hurtling down the mountain's slopes and causing a thunderous noise.	Avalanche
The minimum temperatures in early December have been recorded below 5 degrees Celsius for the first time in 14 years, it said.	Cold Wave
স্থানীয় সময় রাত ২:৪৭ মিনিটে, পশ্চিম নেপালজুড়ে হঠাৎ একটি কম্পন অনুভূত হয়, ফলে হাজারো মানুষ রাস্তায় বেরিয়ে আসে।	Earthquake
ভোরে পাহাড়ের দিক থেকে ঘন ধোঁয়া ও ছাই উঠতে দেখা যায়, যা আশপাশের গ্রামগুলোতে আতঙ্ক ছড়িয়ে দেয়।	Volcanic Activity

Table 1: Dataset Snippet

The tagging process was carried out manually to ensure high accuracy and contextual relevance. Some data can contain vague dictation of any disaster or multiple disasters. These are handled by the generalized tags: Disaster and Non-Disaster. False data is tagged as Non-disaster. This includes metaphorical statements and general facts and opinions about a disaster but not the ones having direct indication of an ongoing hazard.

Sample	Tag
An ice storm occurs when freezing rain coats surfaces with a layer of ice, often causing hazardous conditions.	Non-Disaster
ক্যাটরিনা গ্রে নামে এক ফিলিপাইন সুন্দরী মেয়ন আগ্নেয়গিরির ঢাল বেয়ে নামা লাভার মত উজ্জ্বল গাউন পড়ে ২০১৮ সালের মিস ইউনিভার্স প্রতিযোগিতা জিতে যান।	Non-Disaster

Table 2: Tagging of False Data

Disaster tag is used for any indication of a hazard but not with a clear specification whether it is natural or man made. This tag is also used for data indicating several disasters at once or highlighting significant losses of assets or lives.

Sample	Tag
Residents are returning cautiously, surveying the damage that seems to stretch for miles.	Disaster
অঞ্চলটির পরিস্থিতির দ্রুত অবনতির খবর পাওয়ার পর বাসিন্দাদের সরিয়ে নেওয়া হচ্ছে।	Disaster
স্বত্বসেবীরা সামগ্রী বিতরণে সাহায্য করছেন, তবে ক্ষতির পরিমাণ অনেক বেশি বলে ধারণা করা হচ্ছে।	Disaster

Table 3: Tagging of Ambiguous Data

The dataset includes a balanced distribution of disaster types, with sufficient examples in each category to support effective training and evaluation of machine learning models [3]. The bilingual nature of the dataset adds a unique dimension, enabling the development of systems that cater to underrepresented languages in NLP research.

B. Data Collection and Preprocessing

During the training of a machine learning model, clean and relevant data are essential for enabling the model to learn meaningful patterns and make accurate predictions. In contrast, noisy, incomplete, or irrelevant data can result in poor generalization, overfitting, and biased outcomes [4]. With this in mind, significant emphasis is placed on ensuring the quality of the training dataset for the proposed model.

The data for this research is collected from various news websites, focusing on articles related to natural disasters. Given the scarcity of publicly available datasets containing adequately structured and relevant disaster-related content, the dataset was manually curated to ensure quality and relevance. Each article was systematically reviewed, and sentences containing explicit or implicit references to disaster events were extracted and annotated. As a result, the resulting dataset is both unique and highly representative of real-world textual information, providing a robust foundation for training and evaluating the proposed framework.

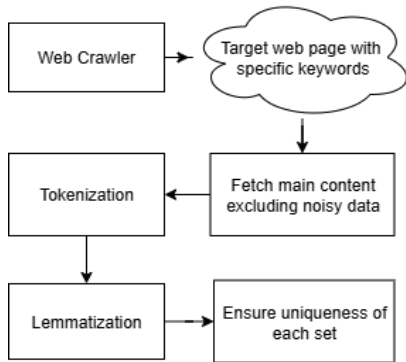


Figure 1: Data Collection and Preprocessing Steps

Figure 1 shows the process of data collection and preprocessing. A web crawler is employed to automatically fetch the content, targeting specific keywords such as names of natural disasters, weather-related events, and disaster management topics. The crawler processes a wide range of news sources to capture diverse perspectives, ensuring a comprehensive dataset. During the data collection process, an essential step involves the removal of irrelevant or noisy content. The crawler is equipped with filters to exclude non-relevant information, such as html tags, navigation contents and spam or advertisement. This preprocessing step maintains the focus on meaningful data, ensuring that only articles containing pertinent information about natural disasters are included.

Punctuation marks, including commas, periods, and quotation marks are retained to ensure the integrity of contextual meaning is maintained as empirical research [5] has underscored the importance of preserving punctuation in syntactic processing. To further refine the dataset, sentence tokenization is implemented within the crawler, enabling the extraction of clean and relevant sentences.

Additionally, duplicate sentences are identified and removed to maintain a dataset of unique entries, reducing redundancy and enhancing clarity. For additional generalization, each entry has been reduced to lowercase form. Lemmatization is also applied to standardize the textual content by reducing words to their base or root forms, ensuring consistency across the dataset. These preprocessing techniques collectively produce a well-structured and comprehensive dataset, optimized for subsequent analysis.

C. Leveraging Transfer Learning

Manually training a deep learning model from scratch for disaster detection would require a vast, well-annotated dataset in both English and Bengali, which is often impractical. Instead, this approach leverages pre-trained contextual embeddings from Bidirectional Encoder Representations from Transformers (BERT), which has already learned rich linguistic patterns across multiple languages. The model is bert-base-multilingual-cased which is a pre-trained transformer model from Hugging Face's Model Hub, that supports 104 languages, including Bengali. BERT's bidirectional architecture allows it to capture the semantic meaning of words in context, making it highly effective for context detection tasks. This significantly improves the model's efficiency and accuracy when processing disaster-related text.

Each sentence of the dataset passes through bert-base-multilingual-cased, which outputs a high-dimensional vector embedding capturing contextual meaning. Formally, given a sequence of words $X = \{w_1, w_2, \dots, w_n\}$, BERT generates contextual embeddings as follows:

$$H = \text{BERT}(X) = \{h_1, h_2, \dots, h_n\}$$

Where X is the input sentence (tokenized into subwords), $BERT(X)$ represents the transformation applied by the pre-trained model, H is the set of hidden state vectors for each token. Each h_i is a contextual representation of the corresponding token w_i .

The process of generating sentence embeddings for the dataset follows these key steps, as illustrated in Figure 2 below.

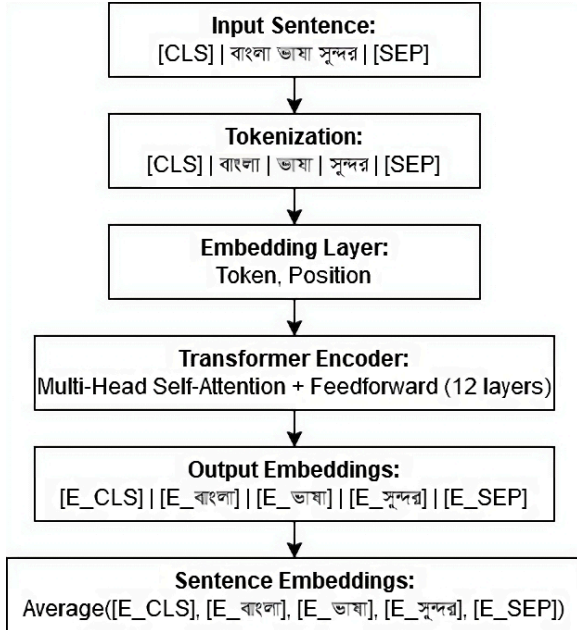


Figure 2: Bidirectional Encoder Representations from Transformers embedding

The raw text is first prepended with [CLS] and appended with [SEP] tokens. Then the sentence is broken down into subword tokens using WordPiece Tokenization to handle unknown or rare words, like punctuations. After that for the Embedding Layer, each token is converted into three types of embeddings: Token (represents the meaning of each token), Position (encodes word order information) and Segment (compares between two sentences) Embeddings. Since the proposed approach works with single sentences, Segment Embeddings are not needed in this case. The token embeddings pass through 12 self-attention layers, where multi-head attention allows each word to be understood in context. This enables the model to recognize disaster-related words based on their surrounding words. The transformer outputs embeddings for each token, where [CLS] contains a summary representation of the entire sentence. To obtain a fixed-size vector representation of the sentence, the average of all token embeddings is taken. This approach ensures that the embeddings capture the overall meaning of the sentence while retaining contextual information.

The use of BERT embeddings offers several advantages, including the ability to handle the morphological richness of Bengali text and the capacity to generalize across diverse sentence structures. These embeddings serve as the

foundation for the similarity-based clustering approach, which is discussed in the next section.

D. Clustering Mechanism

After generating sentence embeddings using BERT, an efficient mechanism is required to store and retrieve similar disaster-related text. For this purpose, Facebook AI Similarity Search (FAISS) has been implemented for high-dimensional clustering and fast retrieval of related sentences. In this approach, FAISS plays a crucial role in structuring disaster-related sentences, ensuring that relevant information can be efficiently retrieved when a new input is provided.

To build the index, each sentence in the dataset—already tagged with one of 18 disaster categories—is first converted into an embedding vector using BERT. Once embedded, each sentence, along with its disaster category, is fed into the index, which organizes the data using approximate nearest neighbor (ANN) search and clusters similar disaster-related text. Instead of manually grouping sentences based on predefined disaster categories, FAISS dynamically clusters similar sentences by placing them in proximity within the vector space. The clustering is performed based on the inherent similarity of the embeddings, allowing the system to generalize across different phrasings and contextual nuances of disaster-related sentences.

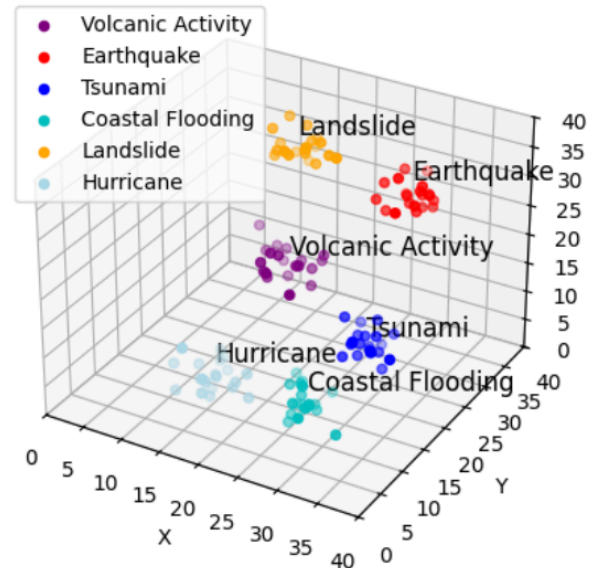


Figure 3: Facebook AI Similarity Search Cluster

Figure 3 shows a sample cluster of disaster tags. The proposed index is constructed using the `FAISS.from_documents()` function, where each sentence embedding, along with its corresponding disaster tag, is altogether stored as a document. When a new query sentence is processed, its predefined embedding is generated and matched against existing indexed embeddings using ANN search. This ensures that

disaster-related sentences with similar contextual meanings are retrieved efficiently. The retrieval mechanism employs a score threshold of 0.8, which filters out low-confidence matches, ensuring that only highly relevant disaster-related content is identified. Mathematically, given an input sentence S , its embedding vector $E(S)$ is computed using the transformer model. This embedding is stored in the FAISS index as $(E(S), \text{Tag}(S))$, where $\text{Tag}(S)$ denotes the disaster category assigned during dataset annotation. During retrieval, a new input sentence Q undergoes the same embedding process to produce $E(Q)$. FAISS then performs an ANN search to find the closest matches $E(S_i)$ within the index, satisfying:

$$\text{Similarity}(E(Q), E(S_i)) \geq \theta$$

where $\theta = 0.8$ is the similarity threshold. The retrieved results provide a set of disaster tags that share contextual meaning with the input query.

IV. RESULTS

A. Dataset Statistics and Configuration

The dataset used in this study comprises a total of 2000 sentences in Bengali, evenly distributed between all 20 categories including Disaster and Non-Disaster, with 100 sentences in each category. Non-Disaster tags are also evenly distributed among categories, such as having each disaster tag's word but metaphorically or having different contextual meaning, thus annotated as Non-Disaster. Similarly, another 2000 sets of data were generated for English corpora, making a total of 4000 unique sets of data. On average, each sentence contains approximately 10 words, although sentence lengths vary depending on the complexity of the reported event and the presence of descriptive or contextual detail.

Tag	Bengali	English	Total
Specific Disaster Tag (eg: Landslide, Earthquake, Hail etc.)	1800	1800	3600
Disaster (general)	100	100	200
Non-Disaster	100	100	200

Table 4: Dataset Distribution

To prepare the data for model training and evaluation, the dataset was divided into training and testing subsets. The training set comprises 80% of the data, while the remaining 20% was reserved for testing purposes. This split ensures that the model is evaluated on unseen examples, providing a

reliable measure of its generalization capability.

B. Evaluation Metrics

The effectiveness of the proposed disaster event detection framework was assessed using widely accepted classification metrics. Precision, defined as the proportion of correctly identified disaster instances among all instances predicted as disasters, measures the model's ability to minimize false positives. Recall, calculated as the proportion of correctly identified disaster instances among all actual disaster cases, evaluates how well the model captures true disaster events. The F1-score, representing the harmonic mean of precision and recall, provides a single metric that balances both measures. Overall accuracy was also reported to indicate the percentage of correctly classified sentences across all categories, encompassing specific disaster types as well as Non-Disaster entries.

The table below summarizes the evaluation metrics for the Bengali and English corpora, illustrating the performance of the proposed framework across all categories:

Category	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Specific Disaster				
Non-Disaster				
Disaster (general)				
Overall				

Table 5: Model Evaluation Metrics

V. FUTURE SCOPE

While the proposed disaster detection model effectively identifies disaster-related information from textual data, several improvements can further enhance its accuracy and applicability. Enhancing the dataset is a key priority for improving disaster detection accuracy. Incorporating Named Entity Recognition (NER) and Part-of-Speech (POS) tagging can refine contextual understanding by identifying critical entities and linguistic patterns. Additionally, semantic role labeling can improve the model's ability to capture relationships between words, leading to more precise classification.

A key limitation of the current approach is that it processes sentences independently, which may lead to misclassifications if disaster cues are spread across multiple sentences. Integrating segment embeddings—which capture relationships between sentences—can improve the model's

contextual understanding. For instance, an isolated sentence like "আগামীকাল ভারী বৃষ্টিপাত হতে পারে।" may not indicate a disaster, but when paired with "ইতোমধ্যে কয়েকটি গ্রাম প্লাবিত হয়েছে।", the meaning becomes clearer. By leveraging multi-sentence embeddings, the model can detect disasters with higher accuracy.

Additionally, sentiment analysis can be integrated to assess the emotional intensity of disaster-related texts. This can help in contextual prioritization, distinguishing between urgent crisis reports and general discussions. It can also reduce false positives by filtering out metaphorical or non-disaster-related mentions.

Expanding the dataset to include code-mixed text (e.g., Bengali-English) would improve the model's adaptability to real-world scenarios. Future work can also explore semi-supervised or active learning to efficiently label new data, reducing reliance on manual annotation. Also, integrating multi-modal data sources like satellite imagery and real-time sensor data can validate and enrich textual information, strengthening disaster response strategies.

These enhancements will further refine the system, making it more robust for real-time disaster detection and response planning.

VI. CONCLUSION

This study presents a novel approach to disaster event detection from unstructured English and Bengali text, leveraging the strengths of deep contextual embeddings and vector-based clustering. The proposed system demonstrates that even for a low-resource language such as Bengali, meaningful disaster-related insights can be extracted effectively when appropriate preprocessing, representation, and semantic embedding strategies are applied.

By utilizing BERT multilingual embeddings and then passing through a FAISS-based similarity search mechanism, the model successfully captures contextual dependencies and clusters text data according to disaster types. Experimental results reveal that the approach achieves balanced precision and recall across categories, indicating its capability to generalize over different forms of disaster narratives.

Overall, this research contributes to the growing field of low-resource language processing by introducing a structured and scalable framework for event detection, paving the way for improved disaster response and information dissemination in Bengali-speaking regions.

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